TRIASHA SARKAR

Atlanta, GA | +1 (404) 948-8761 | tsarkar34@gatech.edu
linkedin.com/in/triasha-sarkar | github.com/triasha72 | triasha72.github.io/Portfolio

Machine Learning Engineer with experience in scientific ML, model evaluation, retrieval, time-series data, and ML systems. Aerospace background with experience in simulation, uncertainty, and engineering problems.

EDUCATION

MS, Aerospace Engineering | Georgia Institute of Technology, Atlanta, GA | Aug 2026

CGPA: 3.7/4.0 | Entered PhD program; transitioned to MS and completed the degree in Aug 2026

MSc, Aerospace Vehicle Design | Cranfield University, United Kingdom | Feb 2023

CGPA: 3.6/4.0 | First Class Honours | Thesis on geometry optimization under FAA constraints

B.Tech, Aerospace Engineering | SRM Institute of Science and Technology, India | Jun 2021

CGPA: 3.8/4.0 | First Class Honours

TECHNICAL SKILLS

ML and scientific ML: PyTorch, PyTorch Geometric, scikit-learn, graph neural networks, neural operators, surrogate modeling, uncertainty quantification, Transformers, RAG, BM25, dense retrieval, reranking

Systems: Python, Go, SQL, Kafka, FastAPI, Docker, Kubernetes, GitHub Actions, PostgreSQL/pgvector, Prometheus, OpenTelemetry, ONNX, Core ML, Qualcomm QNN

Scientific computing: NumPy, SciPy, pandas, CFD meshes, NetworkX, OSMnx, statistical inference, design of experiments

PROFESSIONAL EXPERIENCE

Graduate Research Assistant under Prof. Dimitri Mavris | Georgia Tech ASDL | May 2025 – Aug 2026

- Built and tested Python models for projects in simulation, airline safety, and sustainable aviation; shared reproducible analyses with faculty and sponsors.
- Reviewed assumptions and limits with aerospace experts and documented each workflow so other researchers could rerun it.

Machine Learning Engineer | Rolls-Royce | Jul 2023 – Apr 2025

- Developed Python pipelines to flag unusual engine behavior, support maintenance planning, and combine sensor streams sampled at different rates.
- Mapped certification requirements to data and model checks, then reviewed results with lifecycle engineers before use.

Data Science Intern | Rolls-Royce DataLabs | May 2021 – Jul 2021

- Cleaned aircraft-engine sensor data, built features, and compared prediction and anomaly models; summarized recurring failure patterns for engineers.

TECHNICAL PROJECTS

AIRFAANS | Geometry-aware CFD surrogates | Aug 2026

- Compared a pointwise MLP, MeshGraphNet-style GNN, and point neural operator across three matched seeds and 200 official AirfRANS interpolation meshes per model; MeshGraphNet had the lowest mean field and drag error, while the point operator had the lowest lift error.
- Kept the model explicitly non-operational until Reynolds/AoA OOD, uncertainty, and active-learning experiments are complete; split cases by simulation to prevent node-level leakage.

EdgeGenBench | Real-flight ML and on-device inference | Aug 2026 – Present

- Trained a flight-anomaly model on NASA DASHlink recordings and tested it on 17,780 approaches from held-out aircraft. Its 0.738 macro F1 missed the release target.
- Exported the model to ONNX and tested corrupted sensor inputs; prediction agreement stayed above 99.55% across the tested cases.

NewsLens | Recommendation and real-time search | Jul 2026 – Aug 2026

- Used chronological splits to keep future clicks out of training. The recommender scored 0.366 NDCG@10, but the uncertainty interval still included no improvement.
- Built a Go, Kafka, PostgreSQL, and FastAPI ingestion path for new articles. In a 500-event run, publish p99 was 44 ms and search freshness p95 was 79 ms.

Silent Failure Detection in Rocket-Motor Simulations | Georgia Tech ASDL | May 2026 – Jul 2026

- Traced silent failures in a 15,120-case rocket-motor study to an uncapped convergence loop; a leakage-safe classifier reached 96.8% accuracy on unseen geometries.

Surrogate Model Learning | Reliability under design shift | Feb 2026 – Present

- Evaluated Gaussian-process and conventional surrogates on public airfoil, energy, and concrete data using grouped splits, multi-seed analysis, split-conformal intervals, and distance-to-training-domain guards.
- An airfoil Gaussian process reached R^2 0.8145 on a physically grouped split; a 10-seed mean of 0.8662 ± 0.0680 exposed split sensitivity. Nominal 90% intervals did not retain 90% coverage under design shift.

IntegrityBench | Moderation evaluation and release controls | Aug 2026 – Present

- Built controlled evaluation and release checks for changing moderation policies, keeping separate candidates and frozen benchmark results distinct from deployment claims.

AeroSynth-Eval | Synthetic inspection-image evaluation | Aug 2026 – Present

- Evaluated synthetic aircraft-inspection image data with reproducible dataset splits and reported where model performance was strong, limited, or still pending validation.

Equity Backtest | Walk-forward signal evaluation | Jul 2026 – Present

- Built a cross-sectional equity-signal backtest using walk-forward evaluation, trading costs, and safeguards against common look-ahead and selection biases.

AeroRAG-X | Retrieval, evaluation, and ML systems | Oct 2025 – Present

- Built a technical-knowledge system over 3,233 NASA report sections using hybrid retrieval, reranking, evidence checks, controlled citations, and containerized FastAPI services; retained negative experimental findings rather than claiming every change helped.

GREEN TEA and Project EAGLE | Sustainable aviation modeling | Sep 2025 – Apr 2026

- Built surrogate models to speed up sustainability calculations and connected fuel, demand, transport, and life-cycle inputs; GREEN TEA remains in sponsor use.

HERO | Source-aware retrieval for airline safety | Georgia Tech ASDL | May 2025 – Aug 2026

- Developed retrieval and evaluation methods that link AI-generated airline-safety answers to technical sources analysts can inspect.

Supersonic Business Jets and Noise Mitigation | Cranfield University | Sep 2022 – Feb 2023

- Studied aircraft geometry changes for supersonic business jets, with a focus on noise mitigation, high-speed aerodynamics, and design trade-offs.

Bird-Strike Impact Analysis | SRM Institute of Science and Technology | Jun 2020 – Jun 2021

- Used ANSYS to analyze aircraft nose-cone structures and compare material choices for reducing bird-strike impact.

Ornithopter Aerodynamics | SRM Institute of Science and Technology | Sep 2019 – May 2020

- Studied the structural, mechanical, and aerodynamic behavior of ornithopters; used CFD to examine wing area and load factor.

LEADERSHIP

Graduate Chair | Women of Aeronautics and Astronautics, Georgia Tech | Aug 2025 – Aug 2026

Organizing Team | Aaruush, SRM Institute of Science and Technology | Jul 2019 – Sep 2019

- Helped organize a campus hackathon, gaining early exposure to collaborative, deadline-driven technical projects and the student developer community.

Hackathon Participant | SRM Institute of Science and Technology | Sep 2019

- Worked in a five-member team to plan a sustainable-city modeling project.